

Table 9: Table for the Bit Operators OR, AND and XOR.

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Definition 7: A bit string is a sequence of zero or more bits. The length of this string is the number of bits in the string.

Example 16: Find the bitwise OR, bitwise AND, and bitwise XOR of the bit strings 01 1011 0110 and 11 0001 1101.

$$\begin{array}{r} 01\ 1011\ 0110 \\ 11\ 0001\ 1101 \\ \hline 11\ 1011\ 1111 \end{array}$$

Bitwise OR

$$\begin{array}{r} 01\ 1011\ 0110 \\ 11\ 0001\ 1101 \\ \hline 01\ 0001\ 0100 \end{array}$$

Bitwise AND

$$\begin{array}{r} 01\ 1011\ 0110 \\ 11\ 0001\ 1101 \\ \hline 01\ 0101\ 0101 \end{array}$$

Bitwise XOR